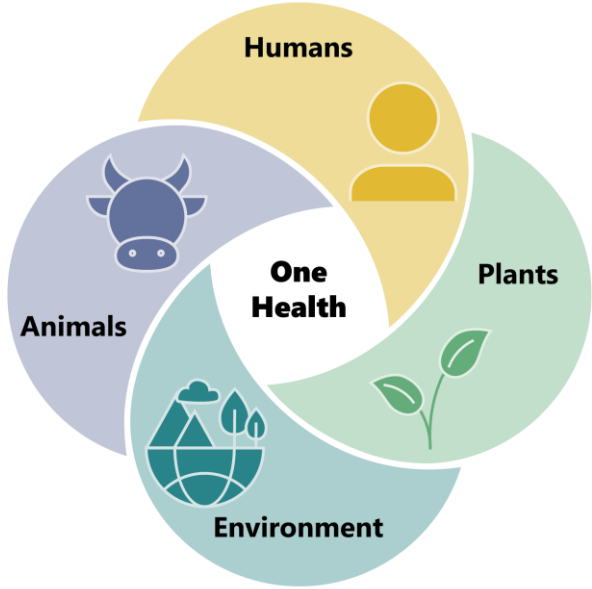
Principal stratum vaccine efficacy with incomplete vaccination

ISBS Conference 2026, Madrid



Florian Klinglmueller, Tobias Fellinger

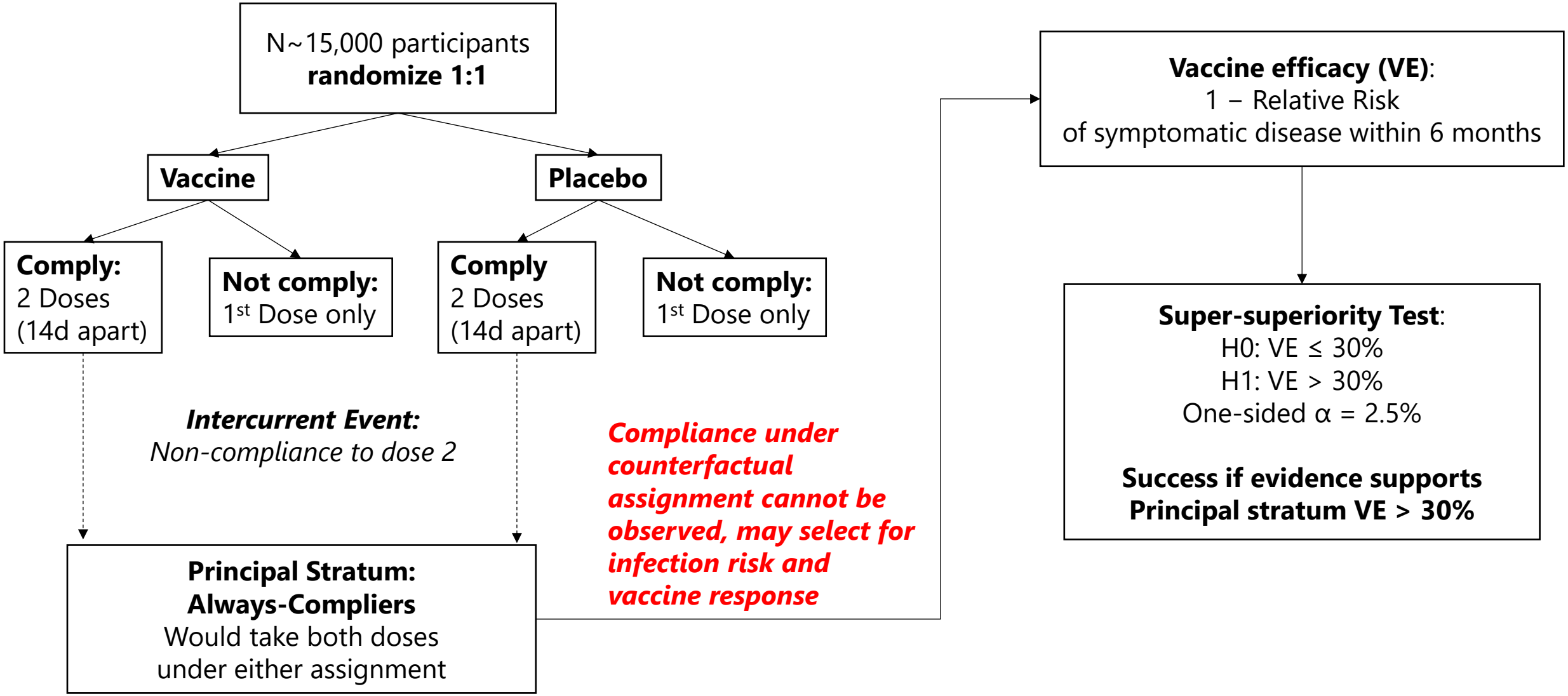
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Design & Estimand

Vaccine efficacy trial for two-dose vaccine with incomplete vaccination



Data generating model

Causal model with prognostic and predictive baseline factors

- Vaccination **A** is randomized and may reduce risk of infection **Y**
- Compliance **C** may depend on and modify the effect of vaccination
- Baseline covariate **V** may impact compliance and risk of infection **Y**
- Baseline covariate **W** may impact compliance, infection risk, and treatment effect

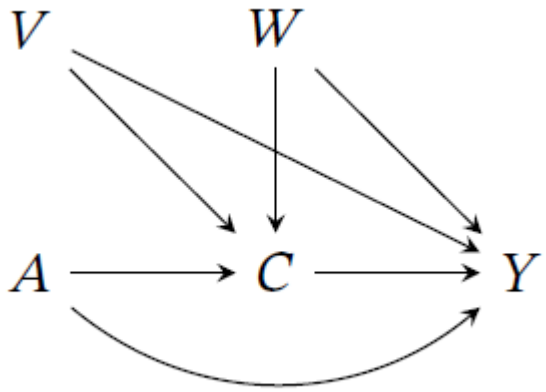


Figure 1: Causal diagram for incomplete vaccination scenario

Description of Scenarios Classes

Short overview of assumptions for data generation

- **Scenario A1**: Compliance is completely at random (~95% receive both doses).
- **Scenario A2**: Vaccine decreases compliance compared to control.
- **Scenario B1**: Vaccine decreases compliance as in A2. In addition, baseline factor V increases compliance, but deteriorate prognosis.
- **Scenario C1**: Vaccine decreases compliance as in A2. Baseline factor W decreases compliance, deteriorates prognosis, but improves protection through active vaccine
- **Scenario D1**: Vaccine decreases compliance as in A2. The study population includes both subjects with baseline factors V and W, conferring the same compliance, prognosis, and treatment effect modification as in B1 and C1

Methods compared

Causal inference approaches and reference method

Method	Label	Assumptions
Instrumental Variable Regression	`iv`	Exclusion restriction, instrument relevance, monotonicity
Principal Score Weighting – adjusted outcome model	`ps_cov`	Correct model specification, principal ignorability, monotonicity
Principal Score Weighting – unadjusted outcome model	`ps_nocov`	
Per-protocol estimation – reference method	`pp`	Compliance unrelated to infection risk and vaccine effect

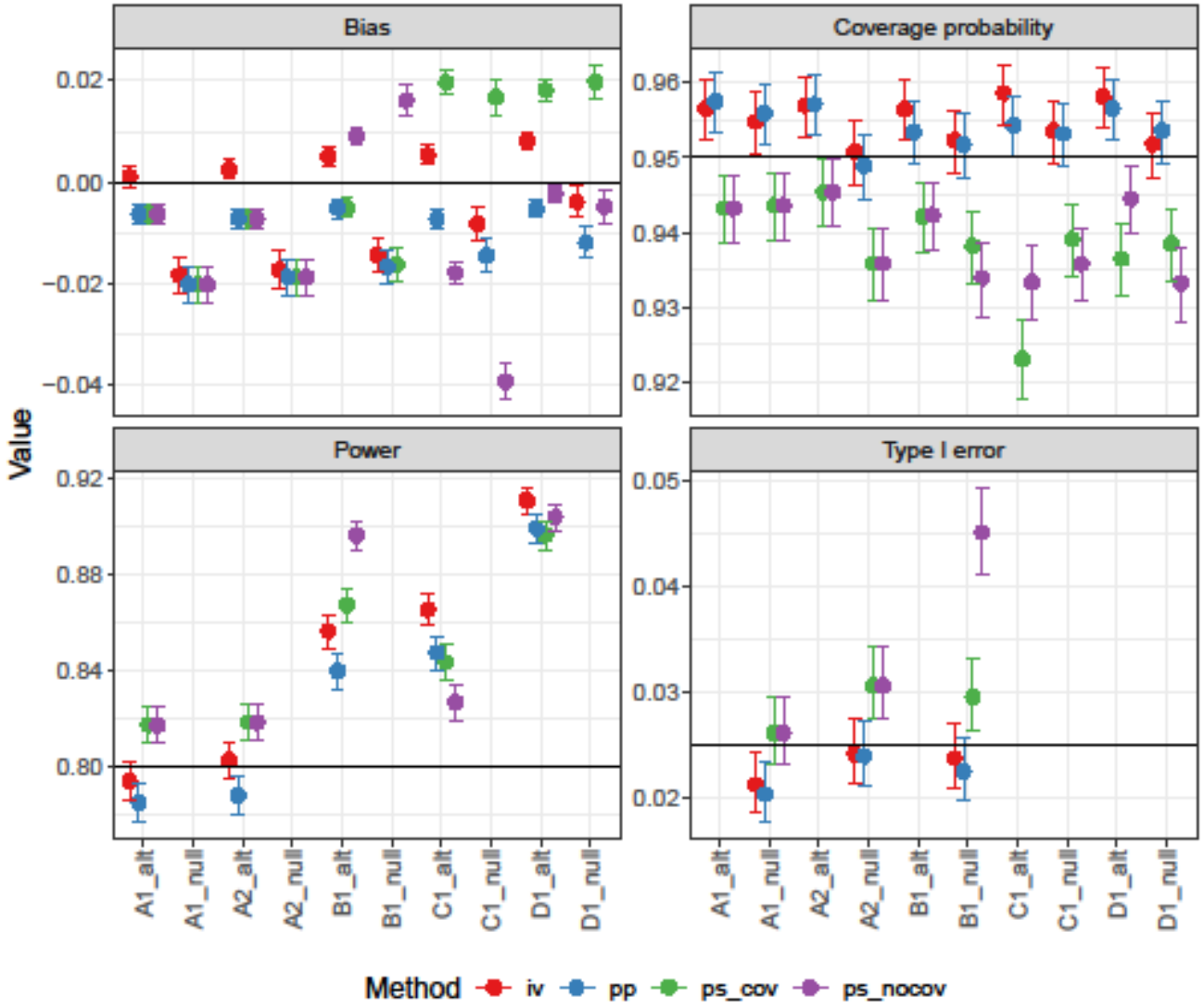
Table 2: Investigated estimation methods for principal stratum estimand

- 10,000 trials simulated per parameter configuration
- PP is a **practically-relevant comparator**, assumption of unrelated compliance is violated in all but the reference scenario
- **Performance measures:** bias, coverage, rejection (type I error / power)

Results for Selected Scenarios & Methods

Modest bias; pp and iv: adequate error control; ps: sensitive to setting

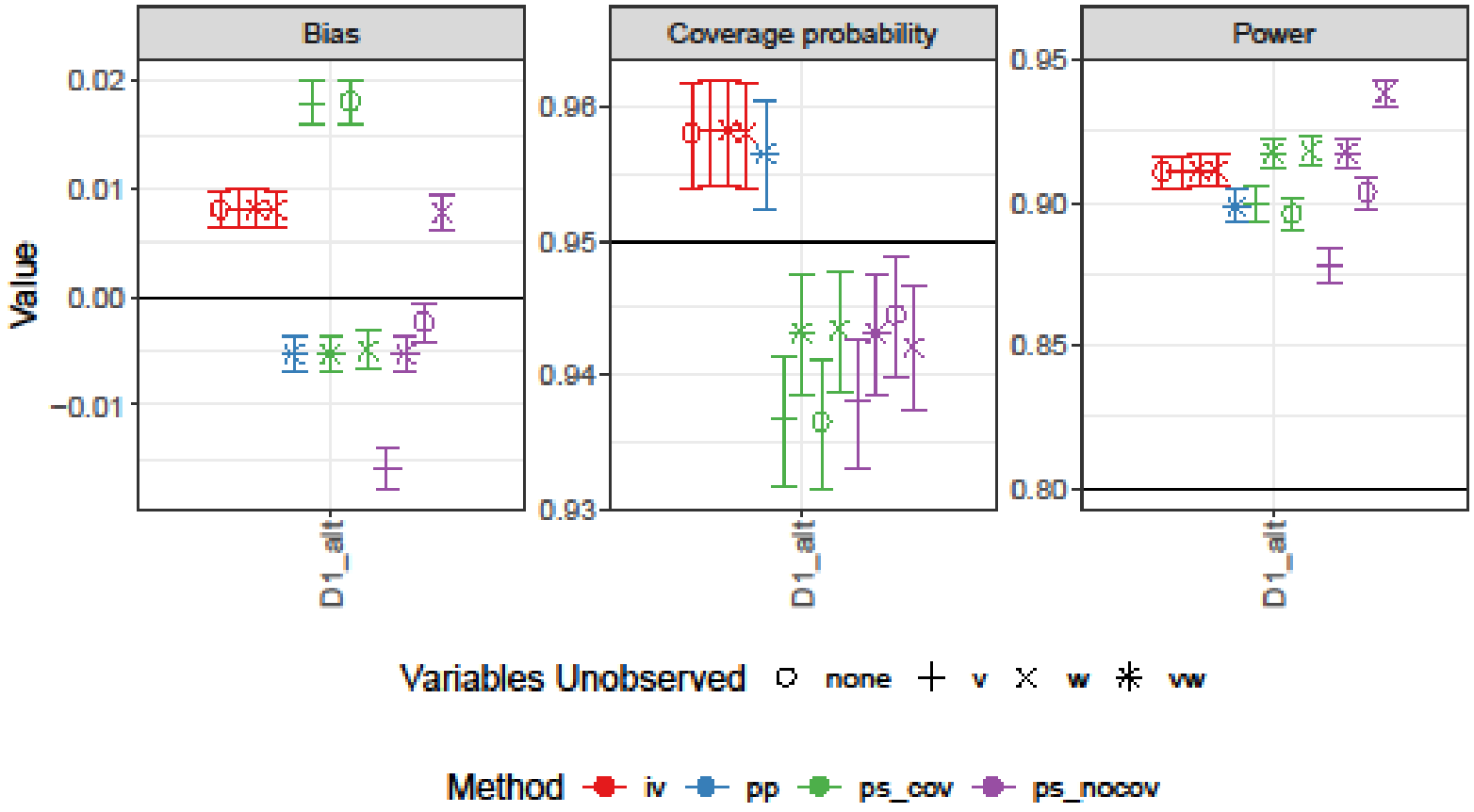
- Moderate bias for all methods up to ~2 percentage points on the VE scale
- **`pp, iv`** control coverage and Type I error
- **`iv`** has slightly higher power and lower bias than the **`pp`**
- **`ps_cov`** or **`ps_nocov`** gave largest bias in all settings
- **`ps_cov`** or **`ps_nocov`** inflated Type I error and had below nominal CI coverage



Results for Observed/Unobserved Covariates

iv: little impact of observation status; ps: sensitive to observation status

- Observation status of baseline factors had little impact on performance for **`iv`**
- Performance of **`ps_cov`**, **`ps_nocov`** varied considerably depending on which factors were observed



Limitations

Extent of violations and factors beside estimand mismatch affect performance

- Transformation to the VE scale introduces negative finite sample bias
- Only few scenarios with larger deviations from assumptions were investigated. E.g. with high non-compliance (50%) performance of the iv estimator deteriorates dramatically
- For several methods, more than one implementation is justifiable, alternative or adjusted methods may be available that could improve the relative performance in this setting.
- Protocol deviations were limited to regimen completion, other deviations and intercurrent events might warrant different analytical treatment

Conclusions

Instrumental variable approach promising; per-protocol estimator competitive

- The instrumental variable approach may be attractive, if non-compliance with dose regimen is limited
- Principal score weighting can improve power and bias in some settings but under-coverage, Type I error inflation, and sensitivity to model specification may limit its applicability
- The per-protocol estimator was often competitive and operationally stable
- While the assumption of unrelated compliance (implied by **pp**) is implausible to hold exactly, violation of the assumptions required for principal scoring methods (**`ps_cov`**, **`ps_nocov`**) may have a larger impact on estimation performance.

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Backup: Parameter values for presented scenarios

Which scenarios are presented in the presentation?

- To best illustrate the effects, scenarios with a high prevalence and impact of the active covariates were included

TABLE C1 | Parameter configurations for vaccine scenario. $\beta_A^{(2)}$ is presented on the VE scale. For both $\beta_A^{(2)}$ and true VE_{AC} both values for the alternative `_alt` and null `_null` settings are given in the same cell separated by a comma. Incidence is the number of infections per person per month corresponding to λ_{post}. Parameters that differ across scenarios are printed in bold.

Scenario	Incidence/month	$e^{\beta_A^{(2)}}$ (_alt,_null)	C	δ_1	γ_A	γ_V	γ_W	γ_{AW}	β_V	β_W	β_{AW}	p_V	p_W	VE_{AC} (_alt,_null)	n_{trt}	n_{ctrl}
A1	1/1000	0.7, 0.3	0.95	T	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0.70, 0.30**	7639	7639
A2	1/1000	0.7, 0.3	0.95	T	**-0.36**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0**	**0.70, 0.30**	7639	7639
B1	1/1000	0.7, 0.3	0.95	T	**-0.36**	**0.5**	**0**	**0**	**0.41**	**0**	**0**	**0.3**	**0**	**0.70, 0.30**	7639	7639
C1	1/1000	0.7, 0.3	0.95	T	**-0.36**	**0**	**-0.8**	**-0.3**	**0**	**0.18**	**-0.22**	**0**	**0.3**	**0.72, 0.34**	7639	7639
D1	1/1000	0.7, 0.3	0.95	T	**-0.36**	**0.5**	**-0.8**	**-0.3**	**0.41**	**0.18**	**-0.22**	**0.3**	**0.3**	**0.72, 0.34**	7639	7639